

Chunlai Long

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Education

Academy of Mathematics and Systems Science, Chinese Academy of Sciences

- Ph.D. in Computational Mathematics 2021.09 - 2026.07 (exp.)
- GPA: 3.73. Coursework: optimization methods, partial differential equations, numerical solutions of differential equations, finite element methods, convex analysis, numerical linear algebra, numerical approximation, stochastic differential equations, parallel computing, inverse problems.

Shihezi University

- B.S. in Mathematics and Applied Mathematics 2017.09 - 2021.06
- GPA: 3.68. Coursework: mathematical analysis, advanced algebra, analytic geometry, abstract algebra, probability and statistics, ordinary differential equations, complex analysis, real analysis, numerical methods, mathematical modeling.

Work Experience

Strategy Algorithm Intern, @Baidu Mobile Ecosystem Data R&D Department

2025.09 - 2025.12

- Reviewed Google DeepMind's AlphaEvolve framework: Program Database, LLM Ensemble, Prompt Sampler, and Evaluator Pool.
- Applied the framework to domain anomaly detection and phishing-site identification; optimized detection models on full-site data and achieved 98% precision in domain anomaly detection.

Research Experience

Multiscale Algorithms for Electromagnetic Models

2020.01 - 2024.01

- Developed HOMO-FDTD for electromagnetic scattering in periodic composites and periodic structures, coupling unit-cell homogenization with time-domain FDTD.
- Reduced grid size and computational cost while preserving physical consistency, making the method suitable for parameter-sensitivity analysis, structural optimization, and large-scale simulations.

Electromagnetic Wave Propagation in Plasma Sheaths

2024.04 - present

- Analyzed Maxwell-Drude and Maxwell-Lorentz coupled systems in dispersive media, focusing on ADE-FDTD schemes with strict stability and convergence requirements.
- Proved temporal-spatial second-order convergence for three ADE-FDTD schemes under regularity assumptions, deriving error estimates and practical criteria for reliable simulations.

Technical Skills

- **Programming and HPC:** C/C++, Python, Linux, MPI, CUDA kernel programming, warp-level parallelism, shared memory management, Grid-Stride Loop.
- **Numerical libraries:** LAPACK/ScaLAPACK, PETSc, BLACS process grids, ScaLAPACK descriptors, DGEQRF, PDGESV, INFO error codes, LWORK/TSIZE workspace queries.
- **Numerical algorithms:** finite element methods, FDTD, ADE-FDTD, homogenization methods, numerical linear algebra, convergence analysis.
- **GPU optimization:** Flash Attention optimization strategies, tiling, register blocking, warp-specialization, streaming and ping-pong scheduling.

Research Publications

- **Convergence Analysis of CPDIM1 Scheme for Maxwell-Drude Model.** In prep, 2025.
- **Convergence Analysis of CPDIM3 Scheme for Maxwell-Drude Model.** In prep, 2025.
- **Convergence Analysis of L-DIM1 Scheme for Maxwell-Lorentz Model.** In prep, 2025.
- **Multiscale Electromagnetic Scattering Simulation via HOMO-FDTD: Coupling Unit-Cell Homogenization with Time-Domain FDTD.** In prep, 2025.

Awards and Certificates

- First Prize, 11th National College Student Mathematics Competition, Mathematics B category.
- First Prize, 12th National College Student Mathematics Competition, Mathematics B category.
- University-level First Prize in Mathematical Modeling Competition; university scholarship.
- CET-4, CET-6, and Teacher Qualification Certificate.